

CORE-ET CPU Subsystem Micro- Architecture Reference

Table of Contents

Document Control	1
Revision History	1
Chapter Status	1
Preface	2
Purpose	2
Audience	2
Sources and Source of Truth	2
Conventions	2
1. ET-SoC-1 Overview	3
1.1. ET-SoC-1 at a Glance	3
1.2. The ET-Minion Hierarchy	3
1.2.1. ET-Minion Core	4
1.2.2. Neighborhood	5
1.2.3. Minion Shire	6
1.2.4. Minion Shire Array	7
1.3. ET-Maxion and Service Processor	7
1.3.1. ET-Maxion Cores	7
1.3.2. Service Processor	8
1.4. Mesh Network-on-Chip	8
1.4.1. Mesh Topology and Mesh Stops	9
1.4.2. Memory Shires	9
1.4.3. PCIe Shire	10
1.4.4. I/O Shire	10

Document Control

Revision History

Chapter Status

Preface

Purpose

Audience

Sources and Source of Truth

Conventions

1. ET-SoC-1 Overview

1.1. ET-SoC-1 at a Glance

ET-SoC-1 is a many-core RISC-V system-on-chip developed primarily for artificial-intelligence and machine-learning (AI/ML) workloads. It is designed as a flexible, fast and efficient inference engine for ML applications. Its compute resources are arranged in a multi-level hierarchy that delivers high sustained performance on regular, parallelizable workloads, such as those of ML algorithms.

The device integrates 1,093 general-purpose 64-bit RISC-V cores of three types:

- 1,088 ET-Minion cores for ML computation
- 4 ET-Maxion cores for management and general-purpose tasks
- 1 Service Processor for boot and device management

The cores are connected to on-die SRAM, external memory and a set of standard interfaces through a mesh network-on-chip (NoC). [Table 1](#) summarizes the device.

Table 1. ET-SoC-1 key characteristics

Item	Description
ET-Minion cores	1,088 in-order, dual-threaded RV64IMFC cores, each with an 8-lane vector/tensor unit
ET-Maxion cores	4 single-threaded, superscalar, out-of-order 64-bit RISC-V cores
Service Processor	1 single-threaded core derived from the ET-Minion
On-die SRAM	140 MB. Each 1 MB block is configurable as L2 cache, as part of the chip-wide L3 cache, or as scratchpad memory.
Interconnect	Mesh NoC with 44 mesh stops
External memory	LPDDR4X, 16 channels × 16 bits, up to 4,266 MT/s
PCI Express	Gen4, 8 lanes. Endpoint, root complex, or both at once.
Other interfaces	eMMC, USB 2.0 OTG, I2C/SMBus, I3C, SPI, UART, GPIO. Debug access through JTAG and a USB 2.0 device port.

1.2. The ET-Minion Hierarchy

The ET-Minion cores are organized in three levels, as shown in [Figure 1](#):

- **ET-Minion core**
- **Neighborhood**: 8 ET-Minion cores
- **Minion Shire**: 4 Neighborhoods (32 ET-Minion cores)

Each level adds resources that are shared by the units it contains. The device contains 34 Minion Shires, which together form the Minion Shire array ([Section 1.2.4](#)).

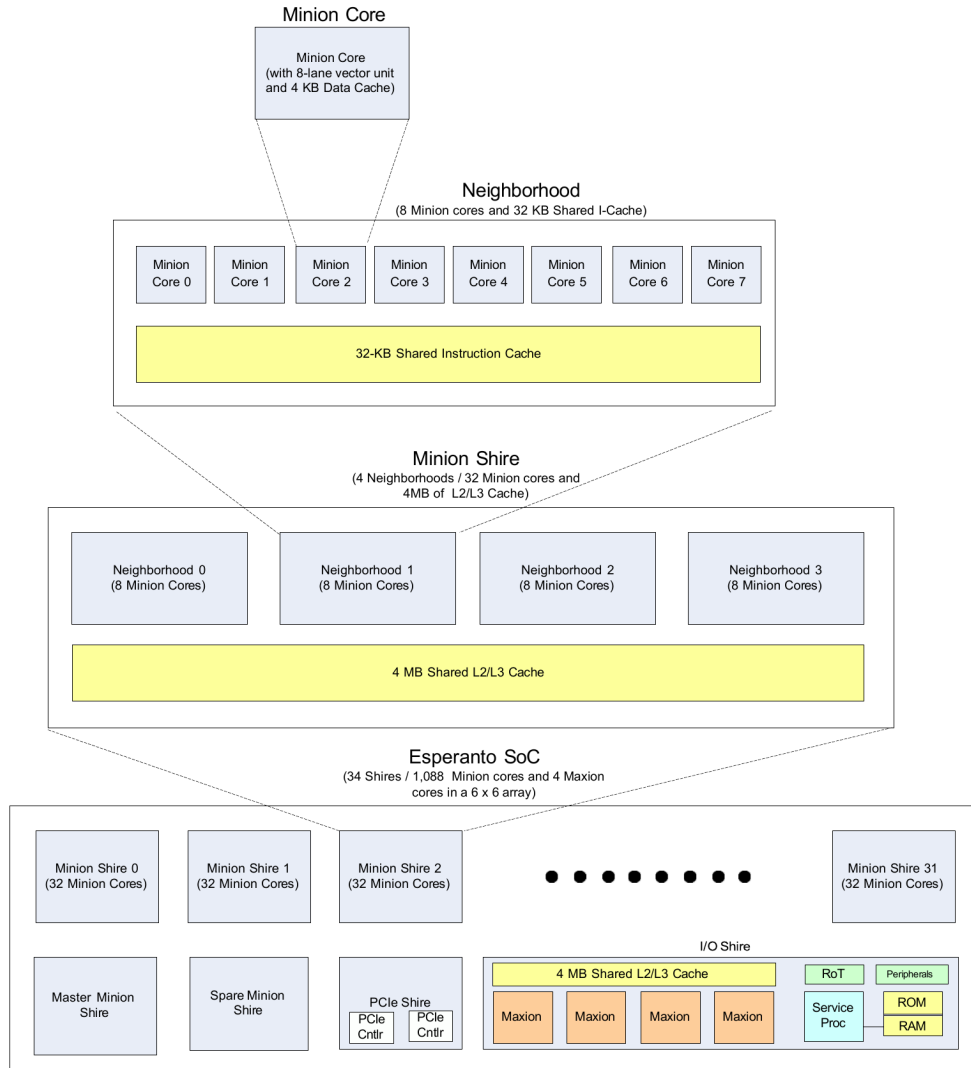


Figure 1. ET-Minion hierarchy

1.2.1. ET-Minion Core

The ET-Minion is a dual-threaded, in-order, single-issue 64-bit RISC-V core that implements RV64IMFC. Its main features are:

- Two hardware threads (harts)
- 4 KB private L1 data cache, partly configurable as scratchpad memory
- Vector processing unit (VPU) with eight identical lanes operating in lock-step. Each lane contains:
 - Floating-point multiply-add unit (FMA): one 32-bit or two 16-bit operations
 - Two integer multiply-add units (IMA): four 8-bit multiply-accumulates each
 - Integer unit (INT)
 - Transcendental unit (TRANS)
- Esperanto instruction extensions for vector, tensor, atomic, message, cache-management and fast-synchronization operations

The core consists of four top-level blocks, shown in [Figure 2](#):

- **Front end (FE):** fetches instructions from the shared instruction cache and decodes them
- **Integer pipeline**
- **L1 data cache and scratchpad**
- **VPU**

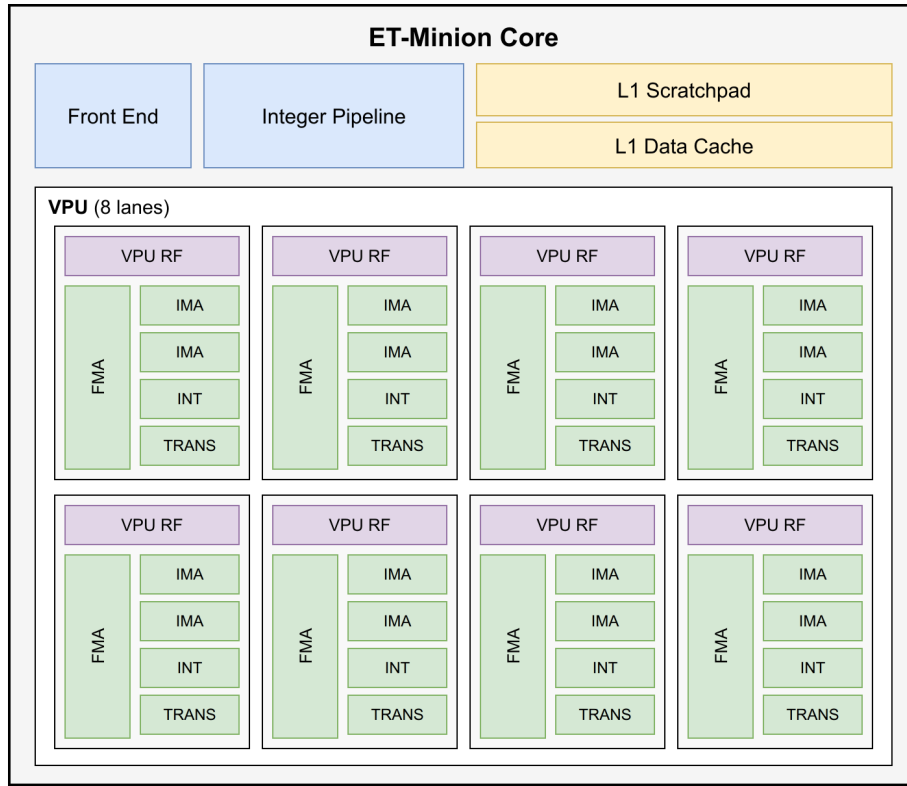


Figure 2. ET-Minion core

1.2.2. Neighborhood

A Neighborhood groups eight ET-Minion cores that share the following resources. In Figure 3, the L0 micro-caches are labeled "I-Cache".

- **L1 instruction cache:** 32 KB, organized as 128 sets $\times$ 4 ways with 64-byte lines. It is shared by all eight cores, and its data RAMs are located outside the Neighborhood.
- **L0 micro-caches:** two, each shared by four cores. Each holds 16 fully associative cache lines, sits between its cores and the L1 instruction cache, and accesses the L1 instruction cache on a miss.
- **Page-table walkers (PTWs):** two, each shared by four cores. Each PTW serves the data caches of its four cores and the instruction-cache side.
- **Request/response path:** a single path to the memory hierarchy outside the Neighborhood. Requests from the instruction and data caches are arbitrated onto it.
- **Performance monitoring unit (PMU):** twelve 64-bit event counters shared by the eight cores. Eight counters count ET-Minion events and four count Neighborhood events.

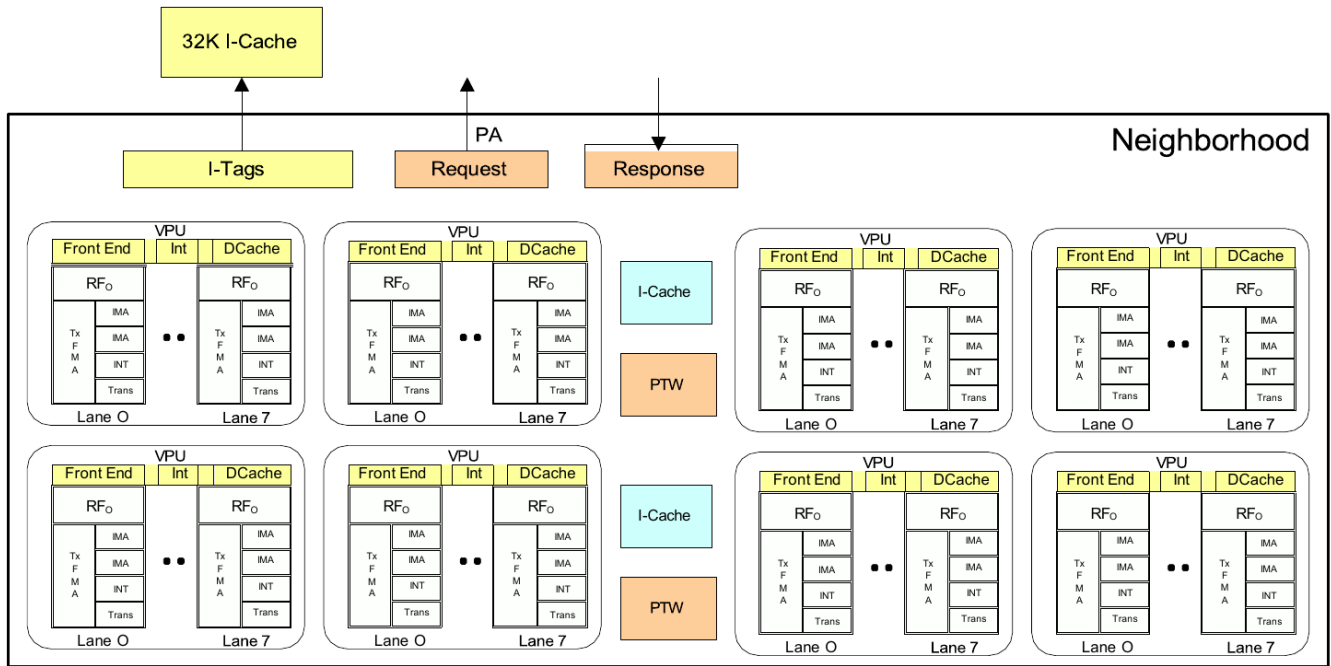


Figure 3. Neighborhood



The PTWs were not implemented in the first silicon revision (A0). The RTL described in this document implements them.

1.2.3. Minion Shire

A Minion Shire groups four Neighborhoods (Figure 4). It also contains:

- **Shire Cache:** 4 MB, in four 1 MB banks, connected to the Neighborhoods through a crossbar. Programmable ESRs divide the cache into three kinds of partition:
 - L2 cache, private to the Shire
 - Slice of the chip-wide distributed L3 cache
 - Scratchpad memory
- **Local services:** uncacheable accesses, fast inter-processor interrupts (IPIs), fast local barriers and fast local credit counters.
- **ESR banks:** the Esperanto System Registers (ESRs) configure the blocks in the Shire.
- **NoC interface:** L2 misses and uncacheable accesses leave the Shire here, bound for other Shires or the memory controllers.

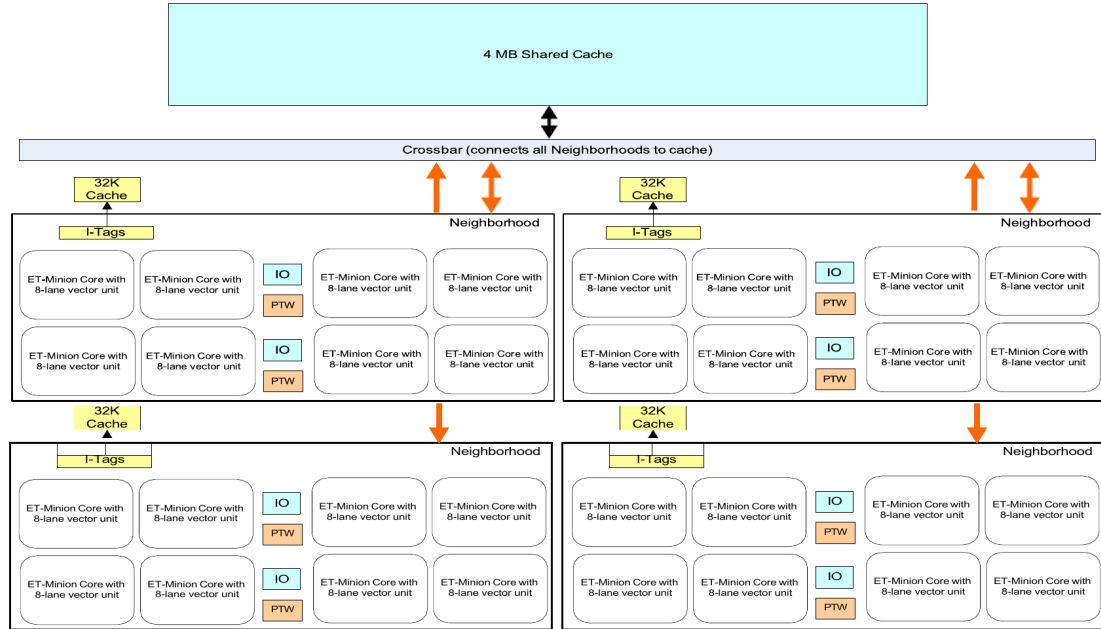


Figure 4. Minion Shire

1.2.4. Minion Shire Array

The 34 Minion Shires, 1,088 ET-Minion cores in total, form the Minion Shire array. The roles of the Shires are assigned by software:

- **Compute:** 32 Minion Shires operate in parallel as the compute array.
- **Master (management):** one Minion Shire receives work and distributes it to the compute array.
- **Spare:** one Minion Shire.

All Minion Shires are identical in hardware, so any Minion Shire can take any of these roles.

1.3. ET-Maxion and Service Processor

In addition to the ET-Minion cores, the device contains two other core types. Both are located in the I/O Shire ([Section 1.4.4](#)). This document does not describe them beyond this section.

1.3.1. ET-Maxion Cores

The ET-Maxion is a single-threaded, superscalar, out-of-order 64-bit RISC-V core intended for management and general-purpose tasks. The device contains four identical ET-Maxion cores. Together with a 4 MB shared L2/L3 cache they form the Maxion Neighborhood, and a system bus and coherency hub in the I/O Shire connects them. [Figure 5](#) shows the ET-Maxion pipeline.

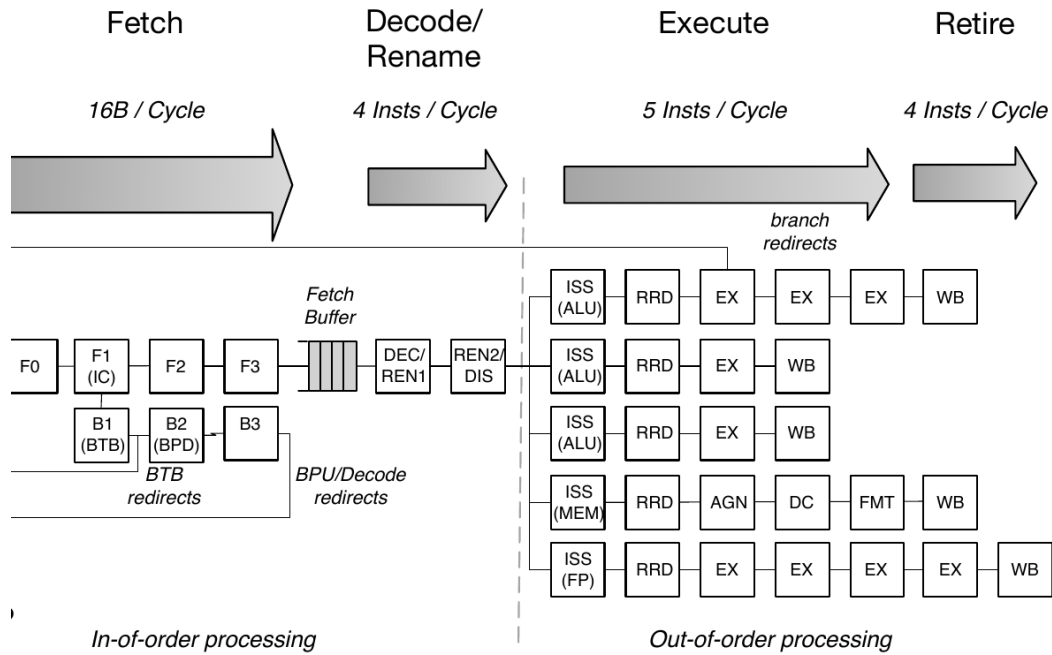


Figure 5. ET-Maxion pipeline

1.3.2. Service Processor

The Service Processor (SP) is a single-threaded ET-Minion core without the vector/tensor extensions. It has its own ROM and a 1 MB scratchpad SRAM, but none of the Neighborhood or Shire shared resources.

- **Boot.** The SP is the first major block released from reset. It executes the boot code from ROM and initializes the device interfaces, including DDR, PCIe and USB.
- **Run time.** It provides management services for the whole device.
- **Access.** The SP can access every resource on the device, and some resources are accessible only to it.

The I/O Shire also contains a hardware root of trust, and six secure mailboxes for communication between four agents:

- Service Processor
- Master Minion Shire
- ET-Maxions
- PCIe host

1.4. Mesh Network-on-Chip

The mesh NoC is the top-level interconnect of ET-SoC-1 and connects all Shires. Figure 6 shows the device with the Shire array on the mesh, the external memory and PCIe interfaces, and the peripheral, Service Processor and debug networks.

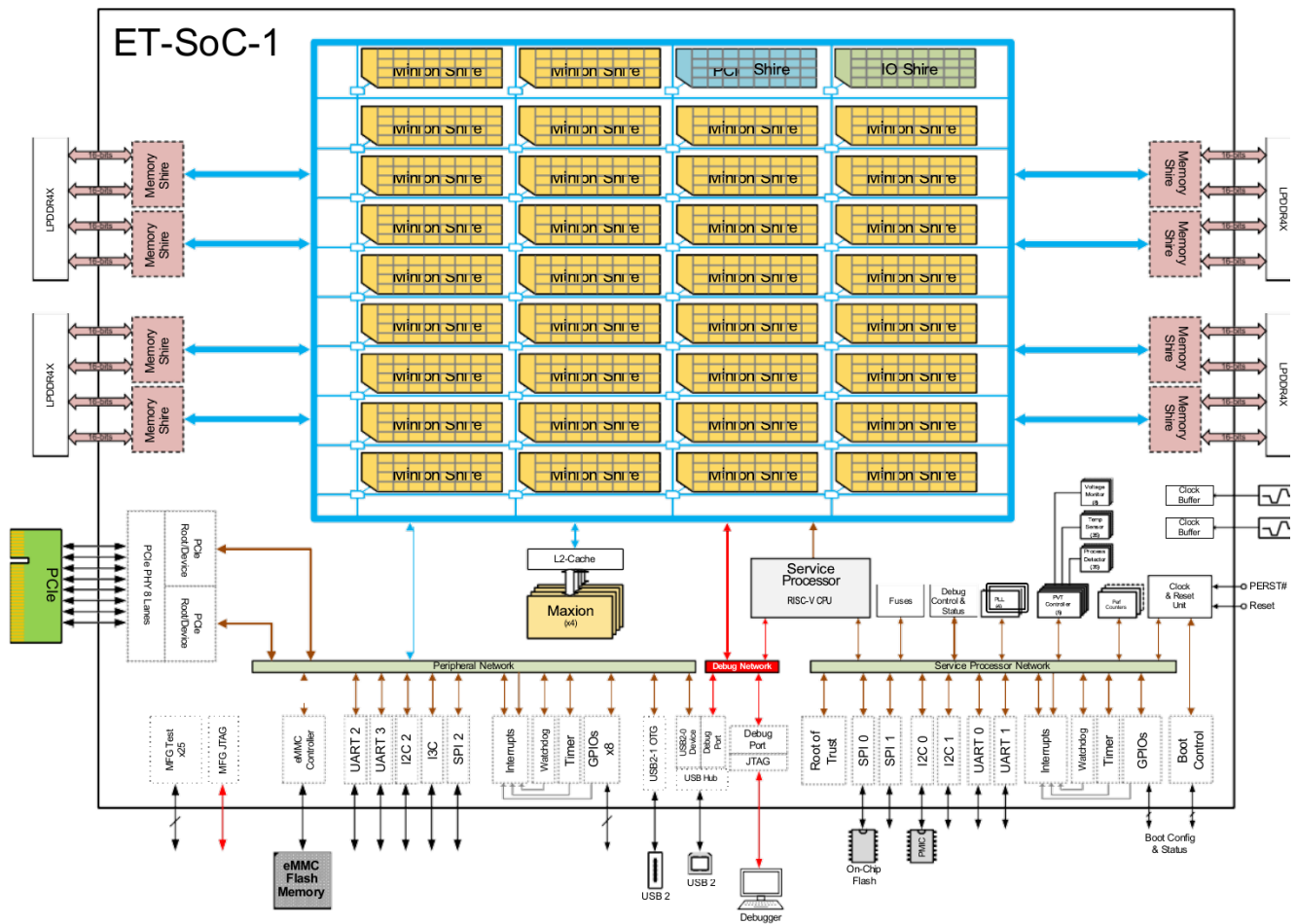


Figure 6. ET-SoC-1 block diagram

1.4.1. Mesh Topology and Mesh Stops

The mesh is organized as an 8×6 grid, and each Shire attaches to it at a mesh stop. The Memory Shires occupy the east and west edges, four on each side, which leaves the corners of the grid unoccupied. The result is 44 mesh stops, listed in Table 2.

Table 2. Shires on the mesh

Shire	Count	Contents
Minion Shire	34	32 ET-Minion cores and a 4 MB Shire Cache
Memory Shire	8	LPDDR4X memory controller
PCIe Shire	1	PCIe controllers and PHY
I/O Shire	1	ET-Maxions, Service Processor and peripherals
Total	44	

1.4.2. Memory Shires

Each of the eight Memory Shires contains one 32-bit LPDDR4X controller that drives two 16-bit channels at up to 4,266 MT/s. The Memory Shires also support global atomic operations.

1.4.3. PCIe Shire

The PCIe Shire contains two dual-mode PCIe Gen4 controllers that share an 8-lane PHY. It supports three lane configurations:

- Single 8-lane endpoint or root complex
- Single 4-lane endpoint or root complex
- 4-lane endpoint together with a 4-lane root complex

The interface is normally factory-configured as an endpoint.

1.4.4. I/O Shire

The I/O Shire contains the ET-Maxions and the Service Processor ([Section 1.3](#)). It also contains the device peripherals, which are grouped on three networks according to which agents may access them.

- **Service Processor network.** Accessible only to the SP. It contains:
 - SPI, including the boot-flash controller
 - I2C, including the link to the external power-management controller
 - UART, including the SP console
 - GPIO, timers, watchdog and boot control
- **Peripheral network.** Shared by the SP, the ET-Maxions and the master Minion Shire. It contains:
 - eMMC, USB 2.0 OTG, I2C/SMBus, I3C, SPI and UART
 - GPIO, timers and watchdog
- **Debug network.** Provides debug access through JTAG and a USB 2.0 device port.